Decision Log

Monday.com Business Intelligence Agent

1. Project Overview

The goal of this project was to design an AI-powered Business Intelligence Agent capable of answering founder-level questions using data from monday.com boards. The agent integrates information from two primary data sources: a Deals board representing the sales pipeline and a Work Orders board representing project execution data.

In real business environments, data is often inconsistent, incomplete, and distributed across multiple systems. This project simulates that scenario by building a system that cleans and analyses messy business data and converts it into meaningful insights.

The agent enables users to query the system conversationally to obtain information about pipeline value, expected revenue, sector performance, and operational metrics.

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2. Key Assumptions

Several assumptions were made while implementing the system:

1. The Deals board represents the company's sales pipeline and includes information such as deal value, probability of closing, deal stage, and sector.
2. The Work Orders board represents executed or ongoing projects and provides operational information such as execution status and billing data.
3. Closure probability values such as "High", "Medium", and "Low" were mapped to numerical probabilities to estimate expected revenue.
4. Missing or incomplete data fields were handled gracefully using data cleaning techniques such as numeric conversion and null value handling.
5. The project assumes that decision-makers primarily require aggregated insights rather than raw transactional data.

These assumptions helped simplify the design while still producing meaningful business insights.

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3. Architecture and Design Decisions

The system was implemented using Python and the Pandas library for data processing and analysis. Google Colab was used as the development environment to ensure easy execution without requiring local setup.

The architecture follows a simple pipeline:

1. Data Ingestion
   The CSV datasets representing monday.com boards are imported and loaded into Pandas DataFrames.
2. Data Cleaning and Normalisation
   Since real-world business data can be inconsistent, several preprocessing steps were performed:

- Converting numeric columns to proper numeric types
- Handling missing values
- Mapping probability categories to numeric scores

3. Business Metrics Calculation
   Key business intelligence metrics were computed, including:

- Total pipeline value
- Expected revenue
- Sector-wise pipeline distribution
- Deal stage distribution
- Project execution status

4. Query Agent Layer
   A lightweight rule-based agent was implemented to interpret user questions and return relevant insights. The agent maps common business questions to the corresponding analytical computations.

This architecture was chosen because it is simple, modular, and easy to extend.

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4. Trade-offs and Design Choices

Several trade-offs were made during development:

Rule-Based Query System vs LLM Integration

A rule-based query system was used to interpret questions rather than integrating a large language model. This decision was made due to time constraints and the requirement to produce a working prototype within a limited timeframe.

However, the architecture can easily be extended to integrate LLM-based query interpretation in the future.

CSV-Based Simulation vs Direct API Integration

The assignment required integration with monday.com boards. For simplicity, CSV exports representing board data were used to simulate the data environment.

With additional development time, the system could directly connect to monday.com using the API for real-time data access.

Simplified Probability Mapping

Probability categories were mapped to fixed numeric values (High = 0.8, Medium = 0.5, Low = 0.2). While this is a simplification, it provides a reasonable estimation for expected revenue calculations.

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5. Handling Data Quality Issues

The dataset included several real-world inconsistencies such as:

- Missing values in numeric fields
- Inconsistent formatting in date columns
- Incomplete deal information

To address these issues, the system implemented:

- Numeric conversion with error handling
- Null value replacement
- Robust aggregation techniques that ignore missing values

The agent also focuses on aggregated insights, which are less sensitive to individual data inconsistencies.

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6. Leadership Update Feature

The optional requirement of preparing data for leadership updates was implemented through an automated report generator. The system produces a summarized business update including:

- Total pipeline value
- Expected revenue
- Top performing sector
- Completed project count

This feature simulates how executives typically receive weekly or monthly performance summaries.

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7. Future Improvements

If additional development time were available, several improvements could be implemented:

1. Direct integration with monday.com APIs for real-time board data access.
2. Integration of large language models for more advanced natural language query understanding.

3. Development of a web-based interface to allow executives to interact with the agent more easily.
4. Implementation of interactive dashboards to visualize business metrics.
5. Advanced forecasting models for revenue prediction.

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8. Conclusion

This project demonstrates how an AI-powered agent can simplify business intelligence workflows by transforming raw operational data into meaningful insights. The system successfully integrates sales pipeline data and operational execution data to answer common leadership questions and generate strategic business summaries.

The modular design ensures that the system can be extended in the future to support real-time integrations, advanced analytics, and more sophisticated natural language interfaces.